

**A First Course in
Design and Analysis
of Experiments**

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of Experiments**

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Chapter 7

Power and Sample Size

The last four chapters have dealt with analyzing experimental results. In this chapter we return to design and consider the issues of choosing and assessing sample sizes. As we know, an experimental design is determined by the units, the treatments, and the assignment mechanism. Once we have chosen a pool of experimental units, decided which treatments to use, and settled on a completely randomized design, the major thing left to decide is the sample sizes for the various treatments. Choice of sample size is important because we want our experiment to be as small as possible to save time and money, but big enough to get the job done. What we need is a way to figure out how large an experiment needs to be to meet our goals; a bigger experiment would be wasteful, and a smaller experiment won't meet our needs.

Decide how large
an experiment is
needed

7.1 Approaches to Sample Size Selection

There are two approaches to specifying our needs from an experiment, and both require that we know something about the system under test to do effective sample size planning. First, we can require that confidence intervals for means or contrasts should be no wider than a specified length. For example, we might require that a confidence interval for the difference in average weight loss under two diets should be no wider than 1 kg. The width of a confidence interval depends on the desired coverage, the error variance, and the sample size, so we must know the error variance at least roughly before we can compute the required sample size. If we have no idea about the size of the error variance, then we cannot say how wide our intervals will be, and we cannot plan an appropriate sample size.

Specify maximum
CI width

Power is
probability of
rejecting a false
null hypothesis

The second approach to sample size selection involves error rates for the fixed level ANOVA F-test. While we prefer to use p -values for analysis, fixed level testing turns out to be a convenient framework for choosing sample size. In a fixed level test, we either reject the null hypothesis or we fail to reject the null hypothesis. If we reject a true null hypothesis, we have made a Type I error, and if we fail to reject a false null hypothesis, we have made a Type II error. The probability of making a Type I error is $\mathcal{E}_I$; $\mathcal{E}_I$ is under our control.

We choose a Type I error rate $\mathcal{E}_I$ (5%, 1%, etc.), and reject H_0 if the p -value is less than $\mathcal{E}_I$. The probability of making a Type II error is $\mathcal{E}_{II}$; the probability of rejecting H_0 when H_0 is false is $1 - \mathcal{E}_{II}$ and is called *power*. The Type II error rate $\mathcal{E}_{II}$ depends on virtually everything: $\mathcal{E}_I$, g , σ^2 , and the α_i 's and n_i 's. Most books use the symbols α and β for the Type I and II error rates. We use $\mathcal{E}$ for error rates, and use subscripts here to distinguish types of errors.

Find minimum
sample size that
gives desired
power

It is more or less true that we can fix all but one of the interrelated parameters and solve for the missing one. For example, we may choose $\mathcal{E}_I$, g , σ^2 , and the α_i 's and n_i and then solve for $1 - \mathcal{E}_{II}$. This is called a power analysis, because we are determining the power of the experiment for the alternative specified by the particular α_i 's. We may also choose $\mathcal{E}_I$, g , $1 - \mathcal{E}_{II}$, σ^2 and the α_i 's and then solve for the sample sizes. This, of course, is called a sample size analysis, because we have specified a required power and now find a sample size that achieves that power. For example, consider a situation with three diets, and $\mathcal{E}_I$ is .05. How large should N be (assuming equal n_i 's) to have a 90% chance of rejecting H_0 when σ^2 is 9 and the treatment mean responses are -7, -5, 3 (α_i 's are -4, -2, and 6)?

Use prior
knowledge of
system

The use of power or sample size analysis begins by deciding on interesting values of the treatment effects and likely ranges for the error variance. "Interesting" values of treatment effects could be anticipated effects, or they could be effects that are of a size to be scientifically significant; in either case, we want to be able to detect interesting effects. For each combination of treatment effects, error variance, sample sizes, and Type I error rate, we may compute the power of the experiment. Sample size computation amounts to repeating this exercise again and again until we find the smallest sample sizes that give us at least as much power as required. Thus what we do is set up a set of circumstances that we would like to detect with a given probability, and then design for those circumstances.

Example 7.1

VOR in ataxia patients

Spinocerebellar ataxias (SCA's) are inherited, degenerative, neurological diseases. Clinical evidence suggests that eye movements and posture are affected by SCA. There are several distinct types of SCA's, and we would like

to determine if the types differ in observable ways that could be used to classify patients and measure the progress of the disease.

We have some preliminary data. One response is the “amplitude of the vestibulo-ocular reflex for 20 deg/s² velocity ramps”; let’s just call it VOR. VOR deals with how your eyes move when trying to focus on a fixed target while you are seated on a chair on a turntable that is rotating increasingly quickly. We have preliminary observations on a total of seventeen patients from SCA groups 1, 5, and 6, with sample sizes 5, 11, and 1. The response appears to have stable variance on the log scale, on which scale the group means of VOR are 2.82, 3.89, and 3.04, and the variance is .075. Thus it looks like the average response (on the original scale) in SCA 5 is about three times that of SCA 1, while the average response of SCA 6 is only about 25% higher than that of SCA 1.

We would like to know the required sample sizes for three criteria. First, 95% confidence intervals for pairwise differences (on the log scale) should be no wider than .5. Second, power should be .99 when testing at the .01 level for two null hypotheses: the null hypothesis that all three SCAs have the same mean VOR, and the null hypothesis that SCA 1 and SCA 6 have the same mean VOR. We must specify the means and error variance to compute power, so we use those from the preliminary data. Note that there is only one subject in SCA 6, so our knowledge there is pretty slim and our computed sample sizes involving SCA 6 will not have a very firm foundation.

7.2 Sample Size for Confidence Intervals

We can compute confidence intervals for means of treatment groups and contrasts between treatment groups. One sample size criterion is to choose the sample sizes so that confidence intervals of interest are no wider than a maximum allowable width W . For the mean of group i , a $1 - \mathcal{E}_I$ confidence interval has width

$$2 t_{\mathcal{E}_I/2, N-g} \sqrt{MSE/n_i} ;$$

for a contrast, the confidence interval has width

$$2 t_{\mathcal{E}_I/2, N-g} \sqrt{MSE} \sqrt{\sum_i \frac{w_i^2}{n_i}} .$$

Width of
confidence
interval

In principle, the required sample size can be found by equating either of these widths with W and solving for the sample sizes. In practice, we don’t

know MS_E until the experiment has been performed, so we must anticipate a reasonable value for MS_E when planning the experiment.

Calculating
sample size

Assuming that we use equal sample sizes $n_i = n$, we find that

$$n \approx \frac{4 t_{\mathcal{E}_I/2, g(n-1)}^2 MS_E \sum w_i^2}{W^2} .$$

This is an approximation because n must be a whole number and the quantity on the right can have a fractional part; what we want is the smallest n such that the left-hand side is at least as big as the right-hand side. The sample size n appears in the degrees of freedom for t on the right-hand side, so we don't have a simple formula for n . We can compute a reasonable lower bound for n by substituting the upper $\mathcal{E}_I/2$ percent point of a normal for $t_{\mathcal{E}_I/2, g(n-1)}^2$. Then increase n from the lower bound until the criterion is met.

If in doubt, design
for largest
plausible MS_E

Often the best we can do is provide a plausible range of values for MS_E . Larger values of MS_E lead to larger sample sizes to meet maximum confidence interval width requires. To play it safe, choose your sample size so that you will meet your goals, even if you encounter the largest plausible MS_E .

Example 7.2

VOR in ataxia patients, continued

Example 7.1 gave a requirement that 95% confidence intervals for pairwise differences should be no wider than .5. The preliminary data had an MS_E of .075, so that is a plausible value for future data. The starting approximation is then

$$n \approx \frac{4 \times 4 \times .075 \times (1^2 + (-1)^2)}{.5^2} = 9.6 ,$$

so we round up to 10 and start there. With a sample size of 10, there are 27 degrees of freedom for error, so we now use $t_{.025, 27} = 2.052$. Feeding in this sample size, we get

$$n \approx \frac{4 \times 2.052^2 \times .075 \times (1 + 1)}{.5^2} = 10.1 ,$$

and we round up to 11. There are now 30 degrees of freedom for error, and $t_{.025, 30} = 2.042$, and

$$n \approx \frac{4 \times 2.042^2 \times .075 \times (1 + 1)}{.5^2} = 10.01 ,$$

so $n = 11$ is the required sample size.

Taking a more conservative approach, we might feel that the MS_E in a future experiment could be as large as .15 (we will see in Chapter 11 that this

is not unlikely). Repeating our sample size calculation with the new MS_E value we get

$$n \approx \frac{4 \times 4 \times .15 \times (1 + 1)}{.5^2} = 19.2 ,$$

or 20 for the first approximation. Because $t_{.025,60} = 2.0003$, the first approximation is the correct sample size.

On the other hand, we might be feeling extremely lucky and think that the MS_E will only be .0375 in the experiment. Repeat the calculation again, and we get

$$n \approx \frac{4 \times 4 \times .0375 \times (1 + 1)}{.5^2} = 4.8 ,$$

or 5 for the first approximation; $t_{.025,12} = 2.18$, so the second guess is

$$n \approx \frac{4 \times 2.18^2 \times .0375 \times (1 + 1)}{.5^2} = 5.7 ,$$

and $n = 6$ works out to be the required sample size.

Note from the example that doubling the assumed MS_E does not quite double the required sample size. This is because changing the sample size also changes the degrees of freedom and thus the percent point of t that we use. This effect is strongest for small sample sizes.

Sample size
affects df and
 t -percent point

7.3 Power and Sample Size for ANOVA

The ANOVA F-statistic is the ratio of the mean square for treatments to the mean square for error. When the null hypothesis is true, the F-statistic follows an F-distribution with degrees of freedom from the two mean squares. We reject the null when the observed F-statistic is larger than the upper $\mathcal{E}_I$ percent point of the F-distribution. When the null hypothesis is false, the F-statistic follows a *noncentral F-distribution*. Power, the probability of rejecting the null when the null is false, is the probability that the F-statistic (which follows a noncentral F-distribution when the alternative is true) exceeds a cutoff based on the usual (central) F distribution.

F-statistic follows
noncentral
F-distribution
when null is false

This is illustrated in Figure 7.1. The thin line gives a typical null distribution for the F-test. The vertical line is at the 5% cutoff point; 5% of the area under the null curve is to the right, and 95% is to the left. This 5% is the Type I error rate, or $\mathcal{E}_I$. The thick curve is the distribution of the F-ratio for one alternative. We would reject the null at the 5% level if our F-statistic is greater than the cutoff. The probability of this happening is the area under

Power computed
with noncentral F

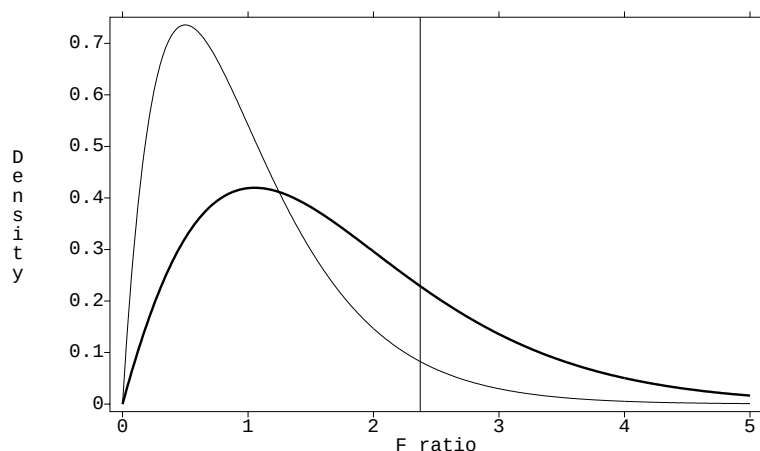


Figure 7.1: Null distribution (thin line) and alternative distribution (thick line) for an F test, with the 5% cutoff marked.

the alternative distribution curve to the right of the cutoff (the power); the area under the alternative curve to the left of the cutoff is the Type II error rate $\mathcal{E}_{II}$.

The noncentral F-distribution has numerator and denominator degrees of freedom the same as the ordinary (central) F, and it also has a *noncentrality parameter* ζ defined by

$$\zeta = \frac{\sum_i n_i \alpha_i^2}{\sigma^2}.$$

The noncentrality parameter measures how far the treatment means are from being equal (α_i^2) relative to the variation of $\bar{y}_{i\bullet}$ (σ^2/n_i). The ordinary central F-distribution has $\zeta = 0$, and the bigger the value of ζ , the more likely we are to reject H_0 .

We must use the noncentral F-distribution when computing power or $\mathcal{E}_{II}$. This wouldn't be too bad, except that there is a different noncentral F-distribution for every noncentrality parameter. Thus there is a different alternative distribution for each value of the noncentrality parameter, and we will only be able to tabulate power for a selection of parameters.

There are two methods available to compute power. The first is to use power tables—figures really—such as Appendix Table D.10, part of which is reproduced here as Figure 7.2. There is a separate figure for each numerator degrees of freedom, with power on the vertical axis and noncentrality param-

Noncentrality
parameter
measures
distance from null

Power curves

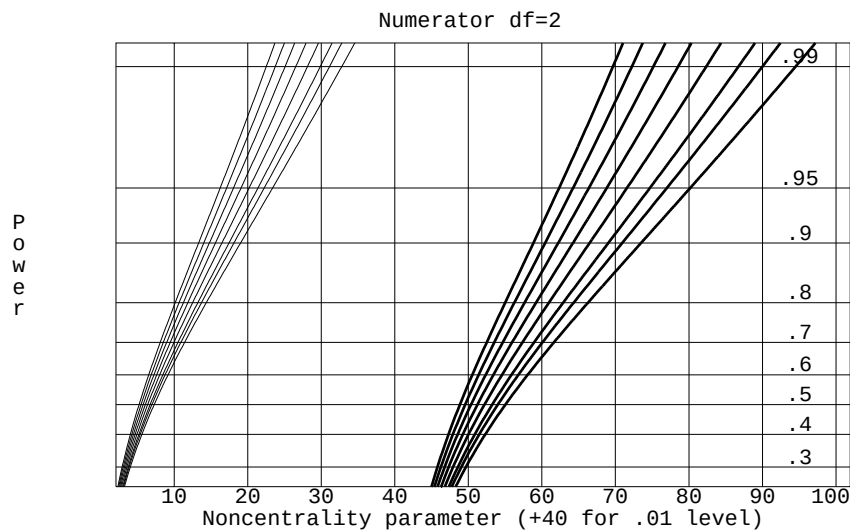


Figure 7.2: Sample power curves for 2 numerator degrees of freedom, .05 (thin) and .01 (thick) Type I error rates, and 8, 9, 10, 12, 15, 20, 30, and 60 denominator degrees of freedom (right to left within each group).

enter on the horizontal axis. Within a figure, each curve shows the power for a particular denominator degrees of freedom (8, 9, 10, 12, 15, 20, 30, 60) and Type I error rate (5% or 1%). The power curves for level .01 are shifted to the right by 40 units to prevent overlap with the .05 curves.

To compute power, you first get the correct figure (according to numerator degrees of freedom); then find the correct horizontal position on the figure (according to the noncentrality parameter, shifted right for .01 tests); then move up to the curve corresponding to the correct denominator degrees of freedom (you may need to interpolate between the values shown); and then read across to get power. Computing minimum sample sizes for a required power is a trial-and-error procedure. We investigate a collection of sample sizes until we find the smallest sample size that yields our required power.

Find required
sample sizes
iteratively

VOR in ataxia patients, continued

We wish to compute the power for a test of the null hypothesis that the mean VOR of the three SCA's are all equal against the alternative that the means are as observed in the preliminary data, when we have four subjects per group and test at the .01 level. On the log scale, the group means in the prelimi-

Example 7.3

nary data were 2.82, 3.89, and 3.04; the variance was .075. The estimated treatment effects (for equal sample sizes) are -.43, .64, and -.21, so the noncentrality parameter we use is $4(.43^2 + .64^2 + .21^2)/.075 = 34.06$. There are 2 and 9 degrees of freedom. Using Figure 7.2, the power is about .92.

Suppose that we wish to find the sample size required to have power .99. Let's try six subjects per group. Then the noncentrality is 51.1, with 2 and 15 degrees of freedom. The power is now above .99 and well off the chart in Figure 7.2. We might be able to reduce the sample size, so let's try five subjects per group. Now the noncentrality is 42.6, with 2 and 12 degrees of freedom. The power is pretty close to .99, but it could be above or below.

Again trying to be conservative, recompute the sample size assuming that the error variance is .15; because we are doubling the variance, we'll double the sample size and use 10 as our first try. The noncentrality is 42.6, with 2 and 27 degrees of freedom. The power is well above .99, so we try reducing the sample size to 9. Now the noncentrality is 38.3, with 2 and 24 degrees of freedom. The power is still above .99, so we try sample size 8. Now the noncentrality is 34.06 with 2 and 21 degrees of freedom. It is difficult to tell from the graph, but the power seems to be less than .99; thus 9 is the required sample size.

Power curves are difficult to use

This example illustrates the major problems with using power curves. Often there is not a curve for the denominator degrees of freedom that we need, and even when there is, reading power off the curves is not very accurate. These power curves are usable, but tedious and somewhat crude, and certain to lead to eyestrain and frustration.

Power software

A better way to compute power or sample size is to use computer software designed for that task. Unfortunately, many statistical systems don't provide power or sample size computations. Thomas and Krebs (1997) review power analysis software available in late 1996. As of summer 1999, they also maintain a Web pagelisting power analysis capabilities and sources for extensions for several dozen packages.¹ Minitab and MacAnova can both compute power and minimum sample size for several situations, including ANOVA problems with equal replication. The user interfaces for power software computations differ dramatically; for example, in Minitab one enters the means, and in MacAnova one enters the noncentrality parameter.

Example 7.4

VOR in ataxia patients, continued

Let's redo the power and sample size computations using Minitab. Listing 7.1 shows Minitab output for the first two computations of Example 7.3. First we

¹<http://sustain.forestry.ubc.ca/cacb/power/review/powrev.html>

Listing 7.1: Minitab output for power and sample size computation.

```

Power and Sample Size                                ①

One-way ANOVA

Sigma = 0.2739  Alpha = 0.01  Number of Levels = 3
Corrected Sum of Squares of Means = 0.6386
Means = 2.82, 3.89, 3.04

Sample
Size    Power
4      0.9297

Power and Sample Size                                ②

One-way ANOVA

Sigma = 0.2739  Alpha = 0.01  Number of Levels = 3
Corrected Sum of Squares of Means = 0.6386
Means = 2.82, 3.89, 3.04

Sample    Target    Actual
Size      Power     Power
5         0.9900    0.9903

```

find the power when we have four subjects per group; this is shown in section ① of the listing. The computed power is almost .93; we read about .92 from the curves. Second, we can find minimum the sample size to get power .99; this is shown in section ② of the listing. The minimum sample size for .99 power is 5, as we had guessed but were not sure about from the tables. The exact power is .9903, so in this case we were actually pretty close using the tables.

Here is a useful trick for choosing sample size. Sometimes it is difficult to specify an interesting alternative completely; that is, we can't specify all the means or effects α_i , but we can say that any configuration of means that has two means that differ by an amount D or more would be interesting. The smallest possible value for the noncentrality parameter when this condition is met is $nD^2/(2\sigma^2)$, corresponding to two means D units apart and all the other means in the middle (with zero α_i 's). If we design for this alternative, then we will have at least as much power for any other alternative with two treatments D units apart.

Specify minimum
difference

7.4 Power and Sample Size for a Contrast

The Analysis of Variance F-test is sensitive to all departures from the null hypothesis of equal treatment means. A contrast is sensitive to particular departures from the null. In some situations, we may be particularly interested in one or two contrasts, and less interested in other contrasts. In that case, we might wish to design our experiment so that the contrasts of particular interest had adequate power.

Noncentrality
parameter for a
contrast

Suppose that we have a contrast with coefficients $\{w_i\}$. Test the null hypothesis that the contrast has expected value zero by using an F-test (the sum of squares for the contrast divided by the MS_E). The F-test has 1 and $N - g$ degrees of freedom and noncentrality parameter

$$\frac{(\sum_{i=1}^g w_i \alpha_i)^2}{\sigma^2 \sum_{i=1}^g w_i^2 / n_i} .$$

We now use power curves or software for 1 numerator degree of freedom to compute power.

Example 7.5

VOR in ataxia patients, continued

Suppose that we are particularly interested in comparing the VOR for SCA 1 to the average VOR for SCA 5 and 6 using a contrast with coefficients (1, -.5, -.5). On the basis of the observed means and MS_E and equal sample sizes, the noncentrality parameter is

$$\frac{(2.82 - .5(3.89 + 3.04))^2}{.075(1/n + .25/n + .25/n)} = 3.698n .$$

The noncentrality parameter for $n = 5$ is 18.49; this would have 1 and 12 degrees of freedom. The power from the tables (testing at .01) is about .86; the exact power is .867.

7.5 More about Units and Measurement Units

Thinking about sample size, cost, and power brings us back to some issues involved in choosing experimental units and measurement units. The basic problems are those of dividing fixed resources (there is never enough money, time, material, etc.) and trying to get the most bang for the buck.

Consider first the situation where there is a fixed amount of experimental material that can be divided into experimental units. In agronomy, the limited

resource might be an agricultural field of a fixed size. In textiles, the limited resource might be a bolt of cloth of fixed size. The problem is choosing into how many units the field or bolt should be divided. Larger units have the advantage that their responses tend to have smaller variance, since these responses are computed from more material. Their disadvantage is that you end up with fewer units to average across. Smaller units have the opposite properties; there are more of them, but they have higher variance.

Subdividing
spatial units

There is usually some positive spatial association between neighboring areas of experimental material. Because of that, the variance of the average of k adjacent spatial units is greater than the variance of the average of k randomly chosen units. (How much greater is very experiment specific.) This greater variance for contiguous blocks implies that randomizing treatments across more little units will lead to smaller variances for treatment averages and comparisons than using fewer big units.

More little units
generally better

There are limits to this splitting, of course. For example, there may be an expensive or time-consuming analytical measurement that must be made on each unit. An upper bound on time or cost thus limits the number of units that can be considered. A second limit comes from edge guard wastage. When units are treated and analyzed *in situ* rather than being physically separated, it is common to exclude from analysis the edge of each unit. This is done because treatments may spill over and have effects on neighboring units; excluding the edge reduces this spillover. The limit arises because as the units become smaller and smaller, more and more of the unit becomes edge, and we eventually we have little analyzable center left.

A second situation occurs when we have experimental units and measurement units. Are we better off taking more measurements on fewer units or fewer measurement on more units? In general, we have more power and shorter confidence intervals if we take fewer measurements on more units. However, this approach may have a higher cost per unit of information.

More units or
measurement
units?

For example, consider an experiment where we wish to study the possible effects of heated animal pens on winter weight gain. Each animal will be a measurement unit, and each pen is an experimental unit. We have g treatments with n pens per treatment ($N = gn$ total pens) and r animals per pen. The cost of the experiment might well be represented as $C_1 + gnC_2 + gnrC_3$. That is, there is a fixed cost, a cost per pen, and a cost per animal. The cost per pen is no doubt very high. Let σ_1^2 be the variation from pen to pen, and let σ_2^2 be the variation from animal to animal. Then the variance of a treatment average is

Costs may vary
by unit type

$$\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{nr}.$$

The question is now, “What values of n and r give us minimal variance of a

treatment average for fixed total cost?” We need to know a great deal about the costs and sources of variation before we can complete the exercise.

7.6 Allocation of Units for Two Special Cases

We have considered computing power and sample size for balanced allocations of units to treatments. Indeed, Chapter 6 gave some compelling reasons for favoring balanced designs. However, there are some situations where unequal sample sizes could increase the power for alternatives of interest. We examine two of these.

Comparison with
control

Suppose that one of the g treatments is a control treatment, say treatment 1, and we are only interested in determining whether the other treatments differ from treatment 1. That is, we wish to compare treatment 2 to control, treatment 3 to control, . . . , treatment g to control, but we don’t compare noncontrol treatments. This is the standard setup where Dunnett’s test is applied. For such an experiment, the control plays a special role (it appears in all contrasts), so it makes sense that we should estimate the control response more precisely by putting more units on the control. In fact, we can show that we should choose group sizes so that the noncontrol treatments sizes (n_t) are equal and the control treatment size (n_c) is about $n_c = n_t \sqrt{g-1}$.

Allocation for
polynomial
contrasts

A second special case occurs when the g treatments correspond to numerical levels or doses. For example, the treatments could correspond to four different temperatures of a reaction vessel, and we can view the differences in responses at the four treatments as linear, quadratic, and cubic temperature effects. If one of these effects is of particular interest, we can allocate units to treatments in such a way to make the standard error for that selected effect small.

Suppose that we believe that the temperature effect, if it is nonzero, is essentially linear with only small nonlinearities. Thus we would be most interested in estimating the linear effect and less interested in estimating the quadratic and cubic effects. In such a situation, we could put more units at the lowest and highest temperatures, thereby decreasing the variance for the linear effect contrast. We would still need to keep some observations in the intermediate groups to estimate quadratic and cubic effects, though we wouldn’t need as many as in the high and low groups since determining curvature is assumed to be of less importance than determining the presence of a linear effect.

Note that we need to exercise some caution. If our assumptions about shape of the response and importance of different contrasts are incorrect, we could wind up with an experiment that is much less informative than the equal

sample size design. For example, suppose we are near the peak of a quadratic response instead of on an essentially linear response. Then the linear contrast (on which we spent all our units to lower its variance) is estimating zero, and the quadratic contrast, which in this case is the one with all the interesting information, has a high variance.

Sample sizes
based on
incorrect
assumptions can
lower power

7.7 Further Reading and Extensions

When the null hypothesis is true, the treatment and error sums of squares are distributed as σ^2 times chi-square distributions. Mathematically, the ratio of two independent chi-squares, each divided by their degrees of freedom, has an F-distribution; thus the F-ratio has an F-distribution when the null is true. When the null hypothesis is false, the error sum of squares still has its chi-square distribution, but the treatment sum of squares has a *noncentral chi-square* distribution. Here we briefly describe the noncentral chi-square.

If $Z_1, Z_2, \dots, Z_n$ are independent normal random variables with mean 0 and variance 1, then $Z_1^2 + Z_2^2 + \dots + Z_n^2$ (a sum of squares) has a chi-square distribution with n degrees of freedom, denoted by χ_n^2 . If the Z_i 's have variance σ^2 , then their sum of squares is distributed as σ^2 times a χ_n^2 . Now suppose that the Z_i 's are independent with means δ_i and variance σ^2 . Then the sum of squares $Z_1^2 + Z_2^2 + \dots + Z_n^2$ has a distribution which is σ^2 times a *noncentral* chi-square distribution with n degrees of freedom and noncentrality parameter $\sum_{i=1}^n \delta_i^2 / \sigma^2$. Let $\chi_n^2(\zeta)$ denote a noncentral chi-square with n degrees of freedom and noncentrality parameter ζ . If the noncentrality parameter is zero, we just have an ordinary chi-square.

In Analysis of Variance, the treatment sum of squares has a distribution that is σ^2 times a noncentral chi-square distribution with $g - 1$ degrees of freedom and noncentrality parameter $\sum_{i=1}^g n_i \alpha_i^2 / \sigma^2$. See Appendix A. The mean square for treatments thus has a distribution

$$MS_{\text{trt}} \sim \frac{\sigma^2}{g-1} \chi_{g-1}^2 \left(\frac{\sum_{i=1}^g n_i \alpha_i^2}{\sigma^2} \right) .$$

The expected value of a noncentral chi-square is the sum of its degrees of freedom and noncentrality parameter, so the expected value of the mean square for treatments is $\sigma^2 + \sum_{i=1}^g n_i \alpha_i^2 / (g - 1)$. When the null is false, the F-ratio is a noncentral chi-square divided by a central chi-square (each divided by its degrees of freedom); this is a noncentral F-distribution, with the noncentrality of the F coming from the noncentrality of the numerator chi-square.

7.8 Problems

- Exercise 7.1** Find the smallest sample size giving power of at least .7 when testing equality of six groups at the .05 level when $\zeta = 4n$.
- Exercise 7.2** We are planning an experiment comparing three fertilizers. We will have six experimental units per fertilizer and will do our test at the 5% level. One of the fertilizers is the standard and the other two are new; the standard fertilizer has an average yield of 10, and we would like to be able to detect the situation when the new fertilizers have average yield 11 each. We expect the error variance to be about 4. What sample size would we need if we want power .9?
- Exercise 7.3** What is the probability of rejecting the null hypothesis when there are four groups, the sum of the squared treatment effects is 6, the error variance is 3, the group sample sizes are 4, and $\mathcal{E}$ is .01?
- Exercise 7.4** I conduct an experiment doing fixed-level testing with $\mathcal{E} = .05$; I know that for a given set of alternatives my power will be .85. True or False?
1. The probability of rejecting the null hypothesis when the null hypothesis is false is .15.
 2. The probability of failing to reject the null hypothesis when the null hypothesis is true is .05.
- Exercise 7.5** We are planning an experiment on the quality of video tape and have purchased 24 tapes, four tapes from each of six types. The six types of tape were 1) brand A high cost, 2) brand A low cost, 3) brand B high cost, 4) brand B low cost, 5) brand C high cost, 6) brand D high cost. Each tape will be recorded with a series of standard test patterns, replayed 10 times, and then replayed an eleventh time into a device that measures the distortion on the tape. The distortion measure is the response, and the tapes will be recorded and replayed in random order. Previous similar tests had an error variance of about .25.
- a) What is the power when testing at the .01 level if the high cost tapes have an average one unit different from the low cost tapes?
 - b) How large should the sample size have been to have a 95% brand A versus brand B confidence interval of no wider than 2?
- Problem 7.1** We are interested in the effects of soy additives to diets on the blood concentration of estradiol in premenopausal women. We have historical data on six subjects, each of whose estradiol concentration was measured at the same stage of the menstrual cycle over two consecutive cycles. On the log scale,

the error variance is about .109. In our experiment, we will have a pretreatment measurement, followed by a treatment, followed by a posttreatment measurement. Our response is the difference (post – pre), so the variance of our response should be about .218. Half the women will receive the soy treatment, and the other half will receive a control treatment.

How large should the sample size be if we want power .9 when testing at the .05 level for the alternative that the soy treatment raises the estradiol concentration 25% (about .22 log units)?

Nondigestible carbohydrates can be used in diet foods, but they may have effects on colonic hydrogen production in humans. We want to test to see if inulin, fructooligosaccharide, and lactulose are equivalent in their hydrogen production. Preliminary data suggest that the treatment means could be about 45, 32, and 60 respectively, with the error variance conservatively estimated at 35. How many subjects do we need to have power .95 for this situation when testing at the $\mathcal{E}_I = .01$ level?

Consider the situation of Exercise 3.5. The data we have appear to depend linearly on delay with no quadratic component. Suppose that the true expected value for the contrast with coefficients (1,-2,1) is 1 (representing a slight amount of curvature) and that the error variance is 60. What sample size would be needed to have power .9 when testing at the .01 level?

Problem 7.2**Problem 7.3**